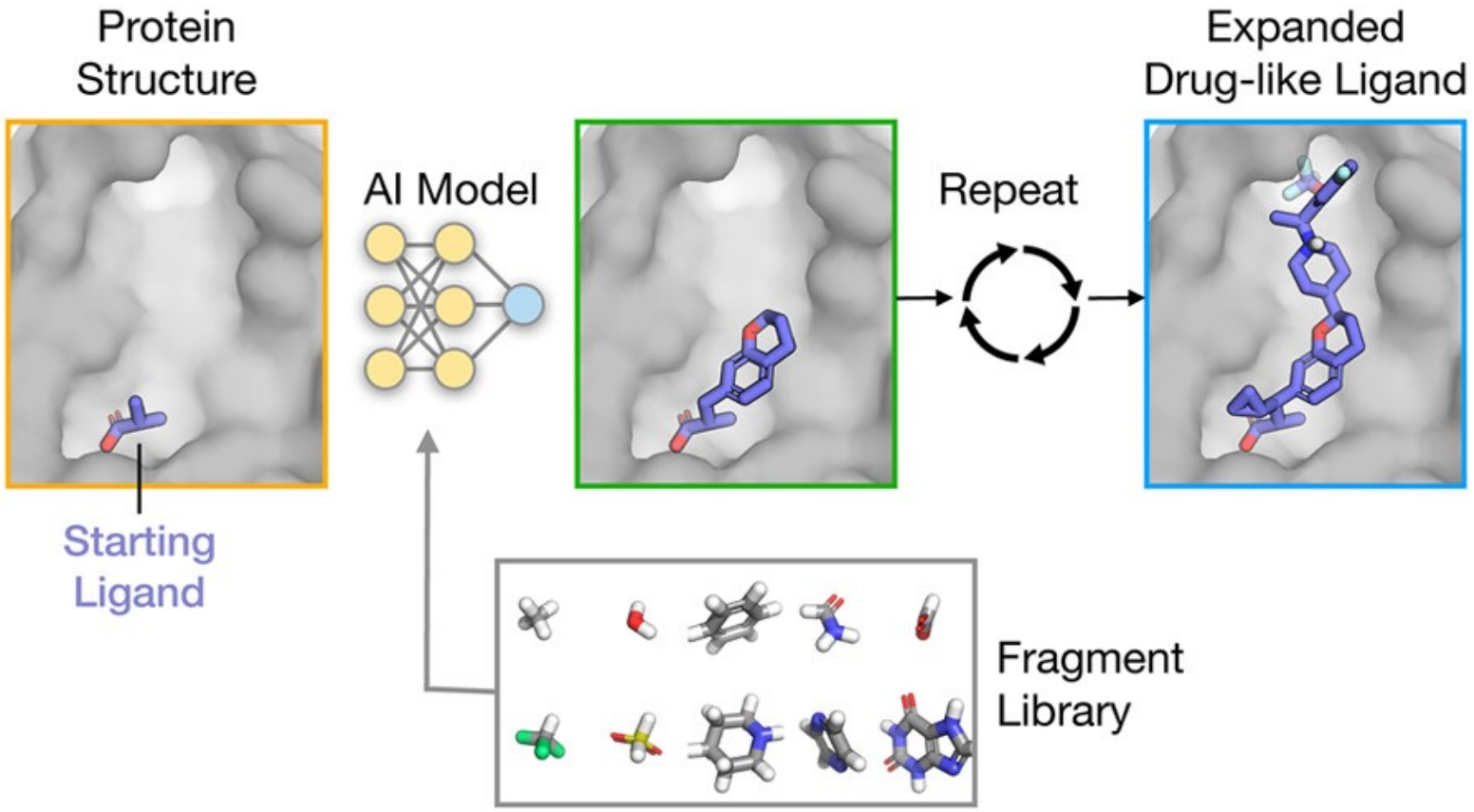
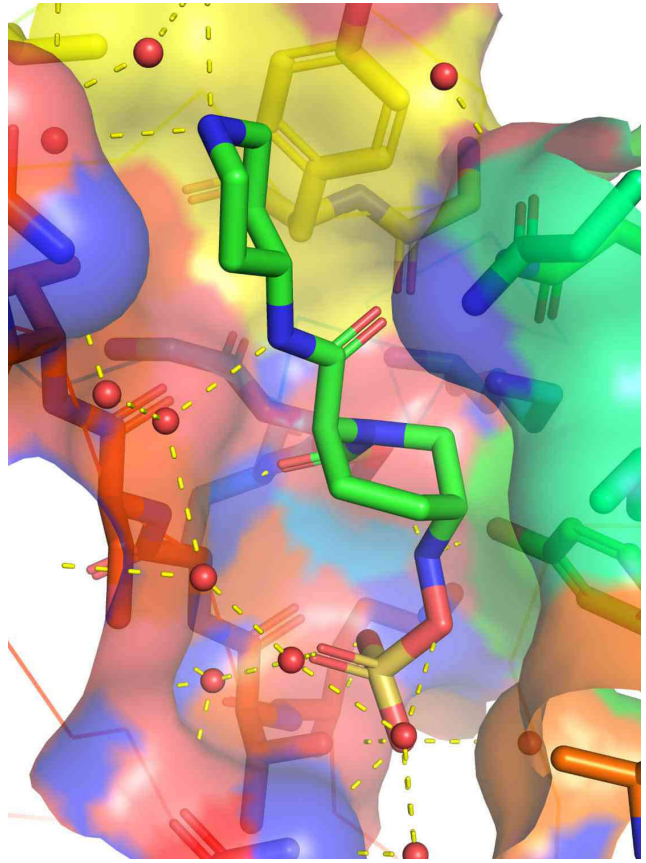
한국과학기술연구원
Korea Institute of Science and Technology

STING agonist design with ChemDesign: Validating Generative Molecular Design Model

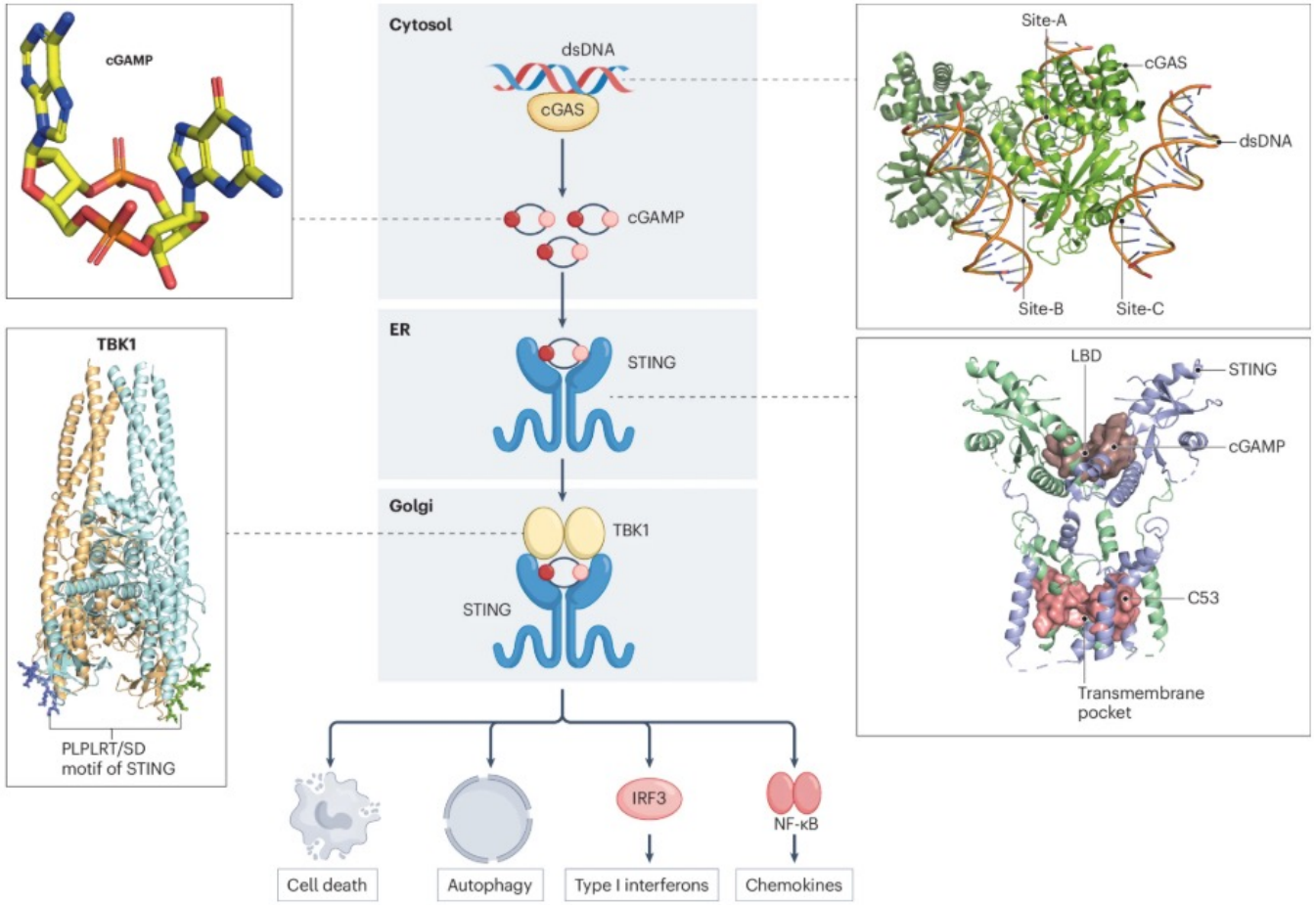
KIST HParkLab
26. 08. 25.
Doyoung LEE

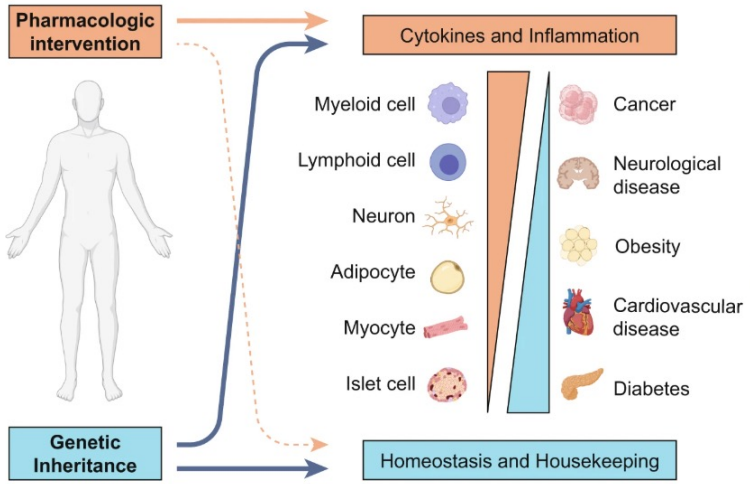
Structure-based drug design





Biological function of STING



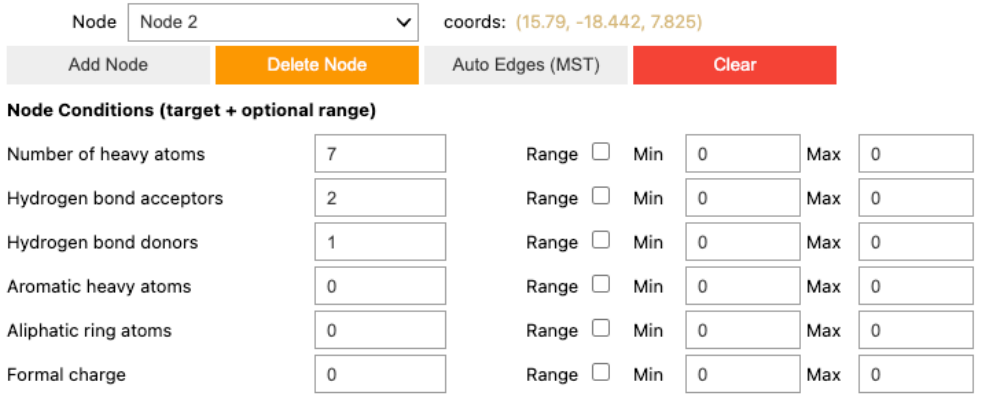


Multifacetated, context-dependent

courtesy of Byunghyun

ChemDesign

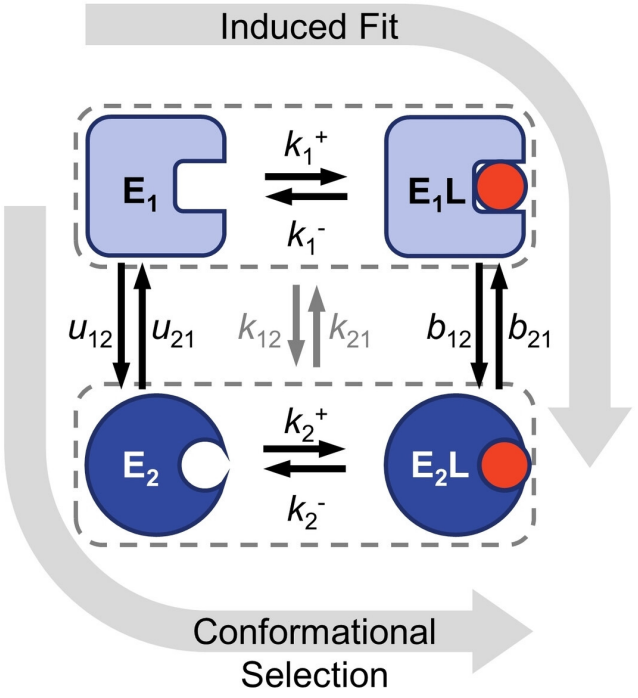




Run	Options	# of samples
1	Fixed fragment	50
2	Vocab similarity (≥ 0.5)	200
3	No similarity	250

courtesy of Byunghyun

ChemDesign



Induced Fit

$$\mathbf{E_1} \underset{k_1^-}{\overset{k_1^+[\mathrm{L}]}{\rightleftharpoons}} \mathbf{E_1 \cdot L} \underset{b_{21}}{\overset{b_{12}}{\rightleftharpoons}} \mathbf{E_2 \cdot L}$$

$$k_{12} = b_{12} \cdot \frac{1}{1 + K_\mathrm{d}/[\mathrm{L}]} \qquad k_{21} = b_{21}$$

$$K_\mathrm{d} = \frac{k_1^-}{k_1^+}$$

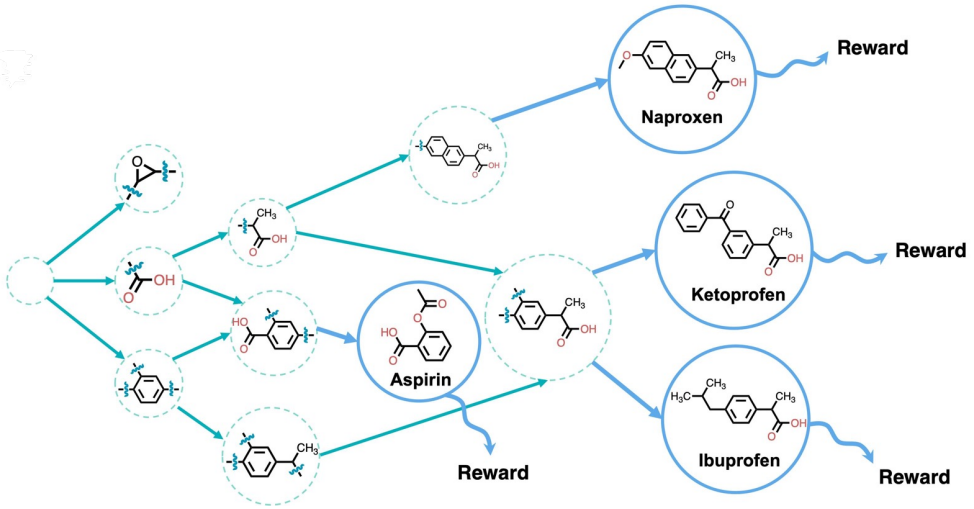
Conformational Selection

$$\mathbf{E_1} \underset{u_{21}}{\overset{u_{12}}{\rightleftharpoons}} \mathbf{E_2} \underset{k_2^-}{\overset{k_2^+[\mathrm{L}]}{\rightleftharpoons}} \mathbf{E_2 \cdot L}$$

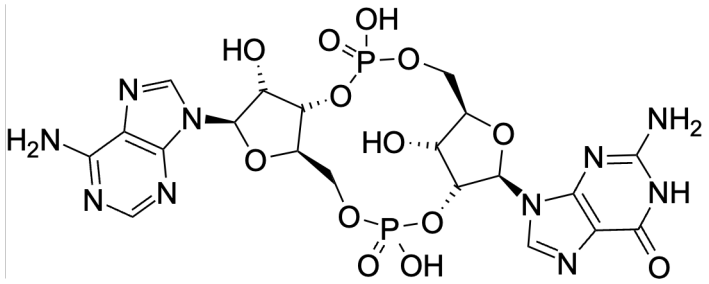
$$k_{12} = u_{12} \qquad k_{21} = u_{21} \cdot \frac{1}{1 + [\mathrm{L}]/K_\mathrm{d}}$$

$$K_\mathrm{d} = \frac{k_2^-}{k_2^+}$$

- Receptor flexibility
- Explicit optimization vs Implicit learning



Reference identification



Endogenous ligand

2', 3'-cGAMP (7SHP)

CDN agonist

non-CDN agonist

Closed conformation
Y167, R238, R232

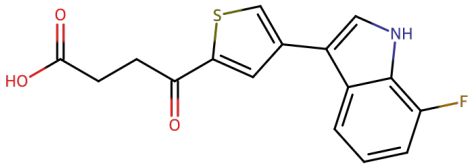
Open conformation
S162, T263, Y240 ···

	Ligand
9IJN	DW18343
4QXP	DMXAA
7SSM	B7L
7X9P	BSP17
8GT6	HB3089
8STH	diABZI-15
9QVT	D5

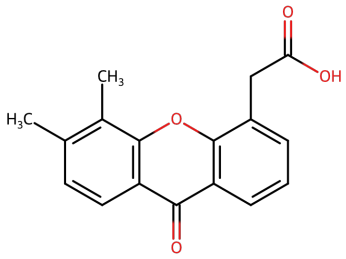
9IJN: Wang X. et al, Med. Comm., 2025
4QXP: Gao P. et al, Cell reports, 2014
7SSM: Cherney E. et al, J. Med. Chem, 2022
7X9P: Feng X. et al, J. Med. Chem, 2022
8GT6: Xie Z. et al, Cell discovery, 2022
8STH: Duvall J.R. et al, J. Med. Chem, 2023
9QVT: Reis C.V. et al, Nat. Chem., 2025

Reference identification

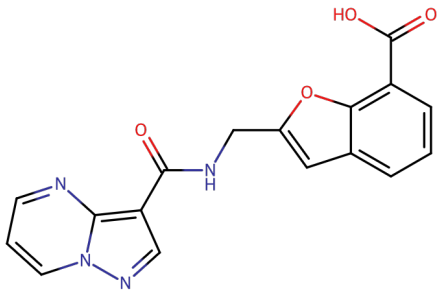
7X9P, 8STH, 9CT3



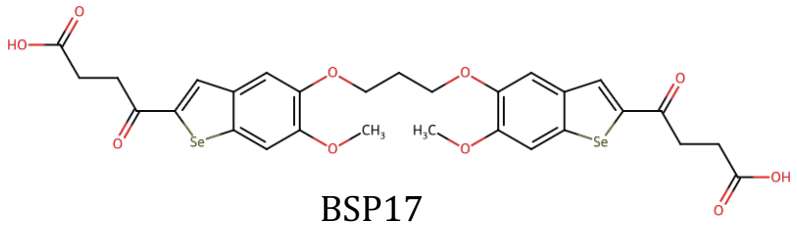
DW18343



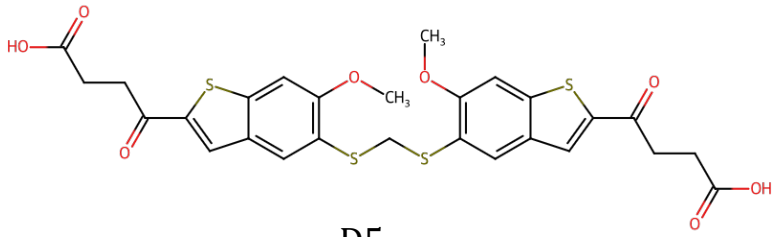
DMXAA



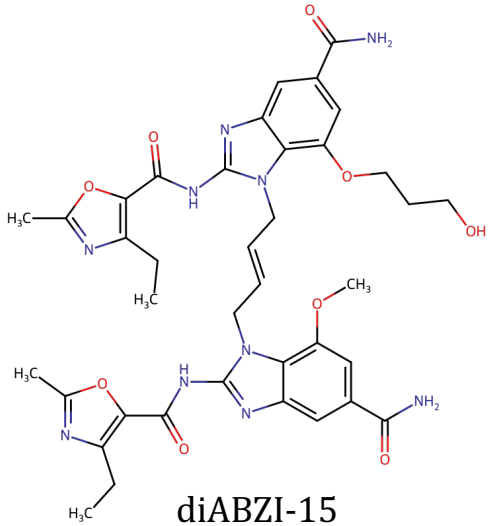
B7L



BSP17



D5

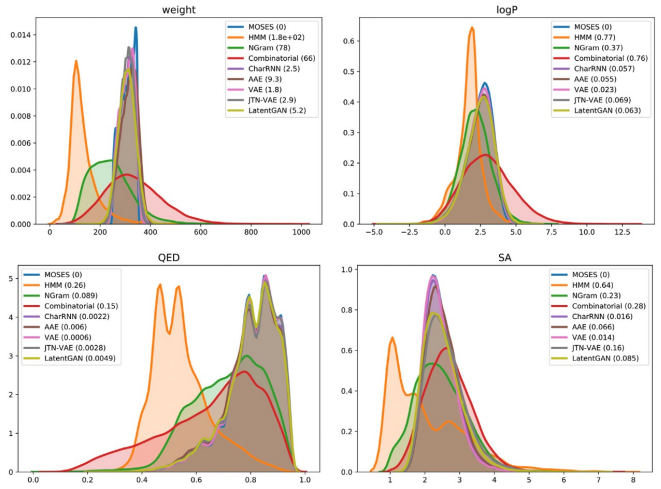


diABZI-15

	Ligand
9IJN	DW18343
4QXP	DMXAA
7SSM	B7L
7X9P	BSP17
8GT6	HB3089
8STH	diABZI-15
9QVT	D5

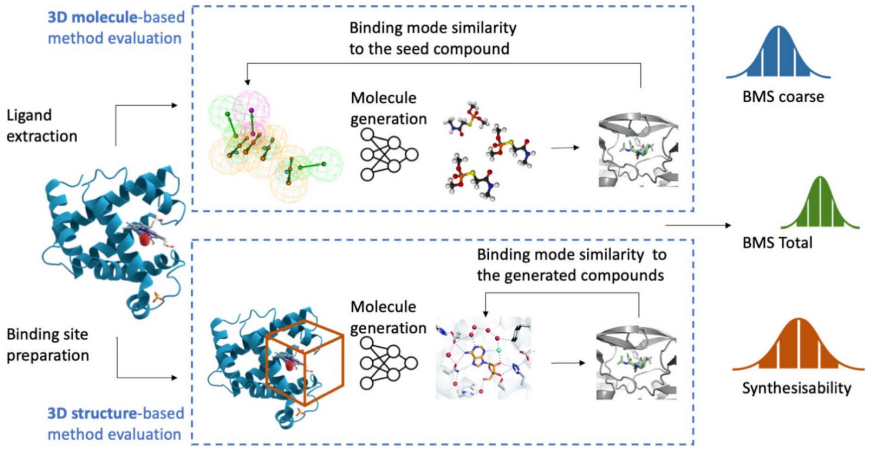
Validation method

Chemical quality (MOSES, GuacaMol) → Redocking → DrugPose approach → AF3 reproduction

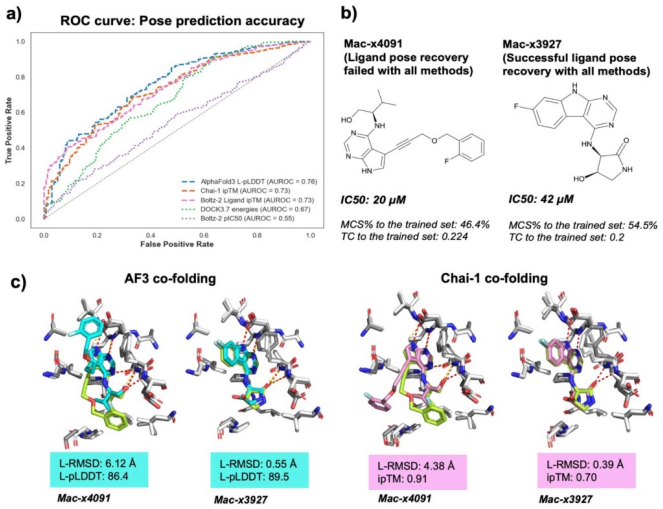


$$\mathrm{FCD}(G,R) = \left\|\mu_G - \mu_R\right\|^2 + \mathrm{Tr}\left[\Sigma_G + \Sigma_R - 2\left(\Sigma_G\Sigma_R\right)^{1/2}\right]$$

Brown et al., 2019 (GuacaMol)
Polykovskiy et al., 2020 (MOSES)

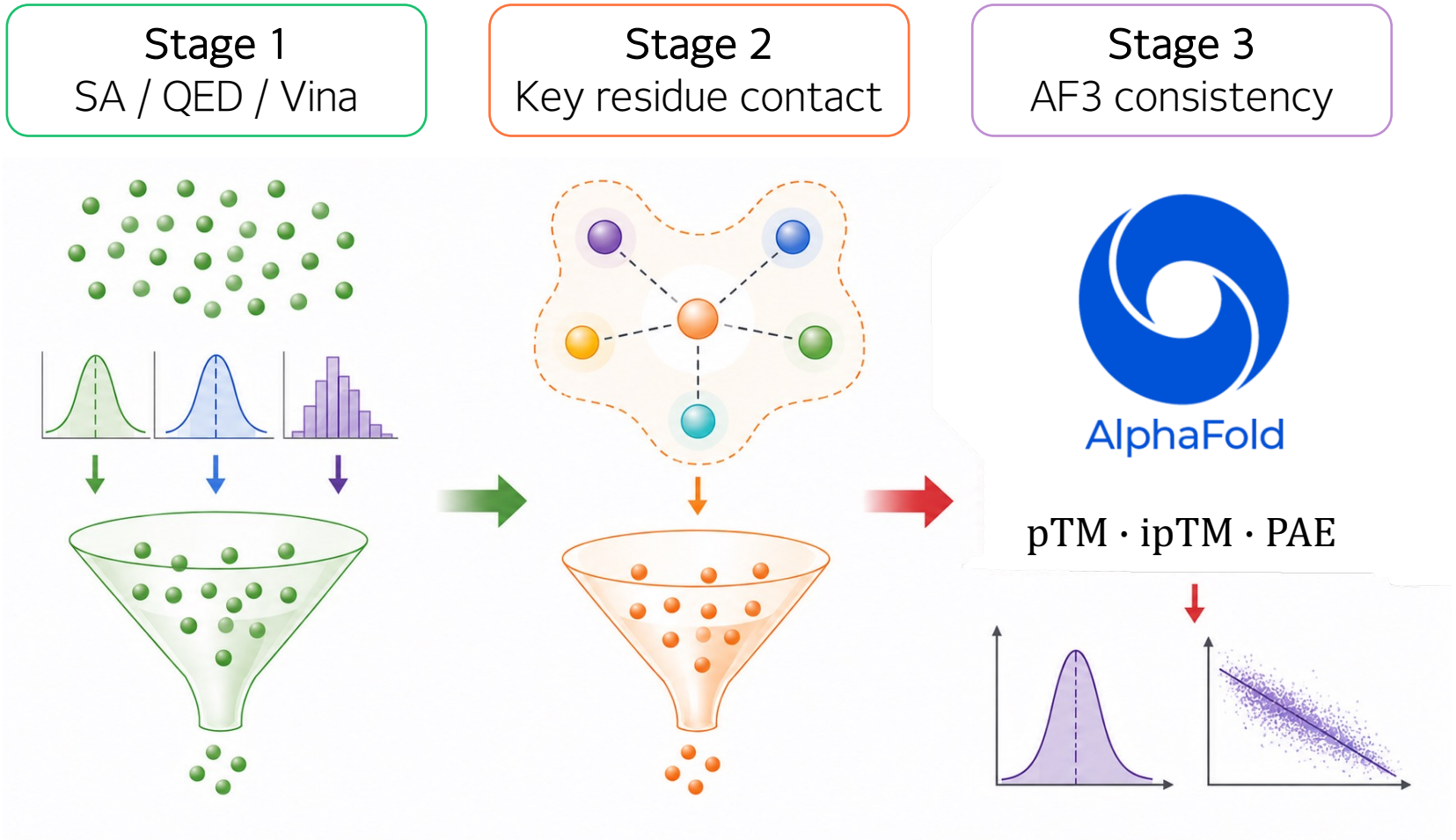


Jocys et al., 2024 (DrugPose)



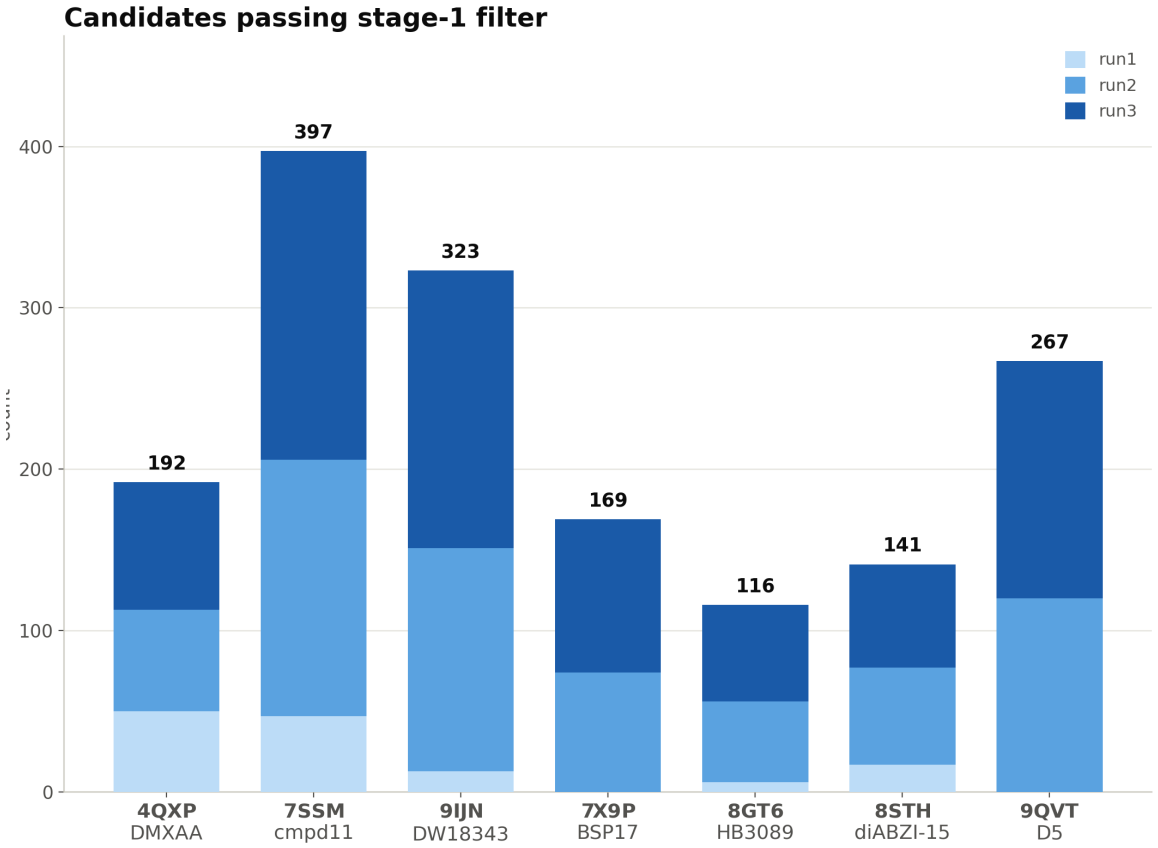
Kim J, et al., 2026

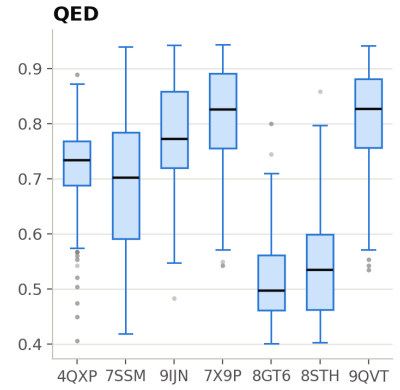
Overall Validation Pipeline

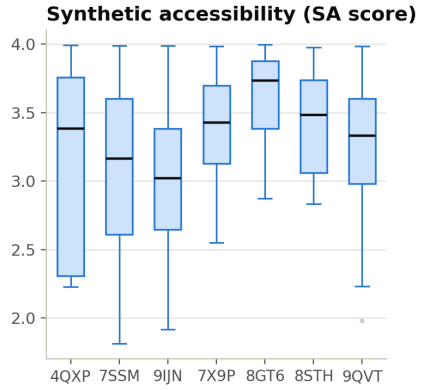


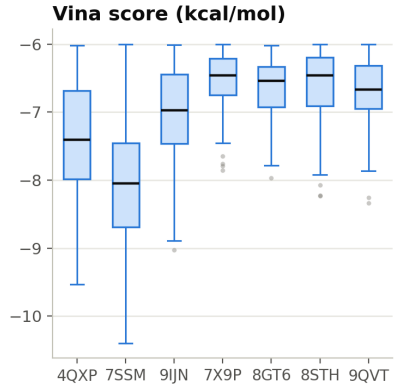
Stage 1: Physicochemical Filter (SA / QED / VINA)

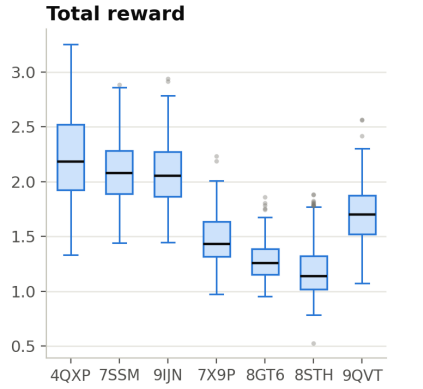
1605 candidates passing across 7 STING pockets (QED ≥ 0.4, SA ≤4.0, Vina score ≤ -6.0)











Stage 2: Key residue contact Filter

Candidates should contact with one of Primary anchor residue (R238, T263, S162)

	Primary anchor Res.	Auxiliary stabilizing Res.
9IJN	R238, Y167, T263	S162, P264
4QXP	R238, T263	T267
7SSM	R232, R238	T263, S162
7X9P	R238, Y167	I235, T263
8GT6	S162, V239, Y240	L159, Y167, T263
8STH	Y167, S162, T263	R238, Y163, Y240
9QVT	R238, R232, T263	Y167

Primary anchor residue

Essential for binding or activation
hard filtering criteria

Auxiliary stabilizing residue

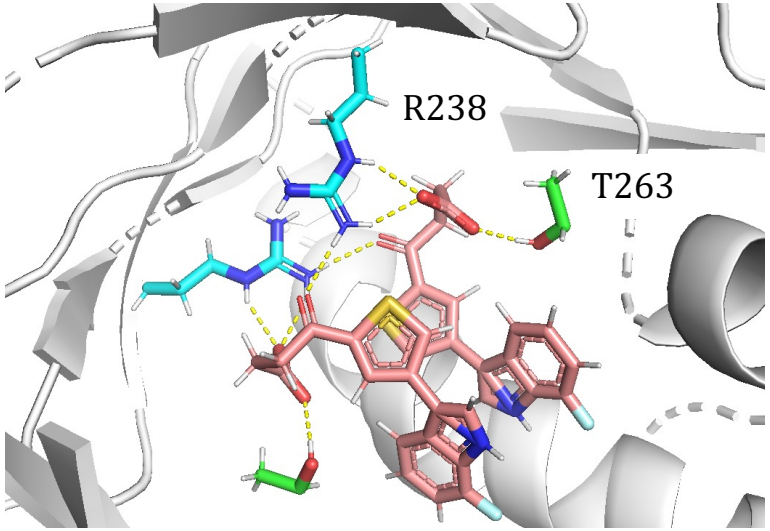
Provide additional binding stability
Potential scoring

Stage 2: Key residue contact Filter

Candidates should contact with one of Primary anchor residue (R238, T263, S162)

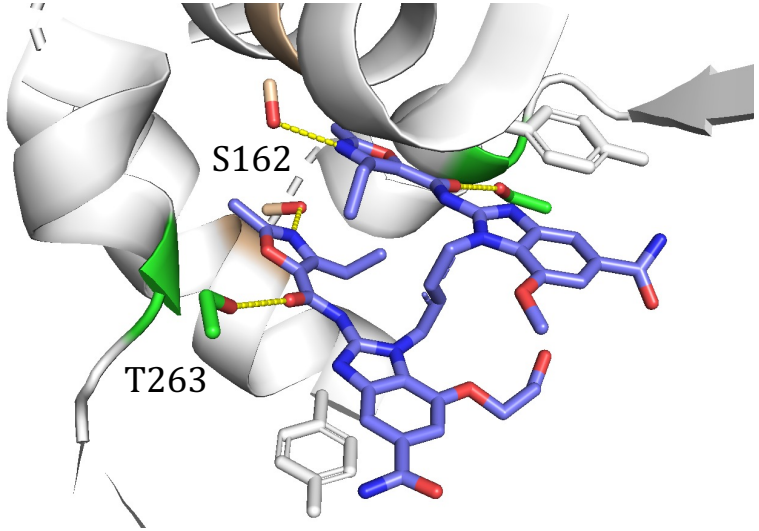
	Primary anchor Res.
9IJN	R238, Y167, T263
4QXP	R238, T263
7SSM	R232, R238
7X9P	R238, Y167
8GT6	S162, V239, Y240
8STH	Y167, S162, T263
9QVT	R238, R232, T263

[Closed conformation] (e.g. 9IJN)



Contact with R238 and T263

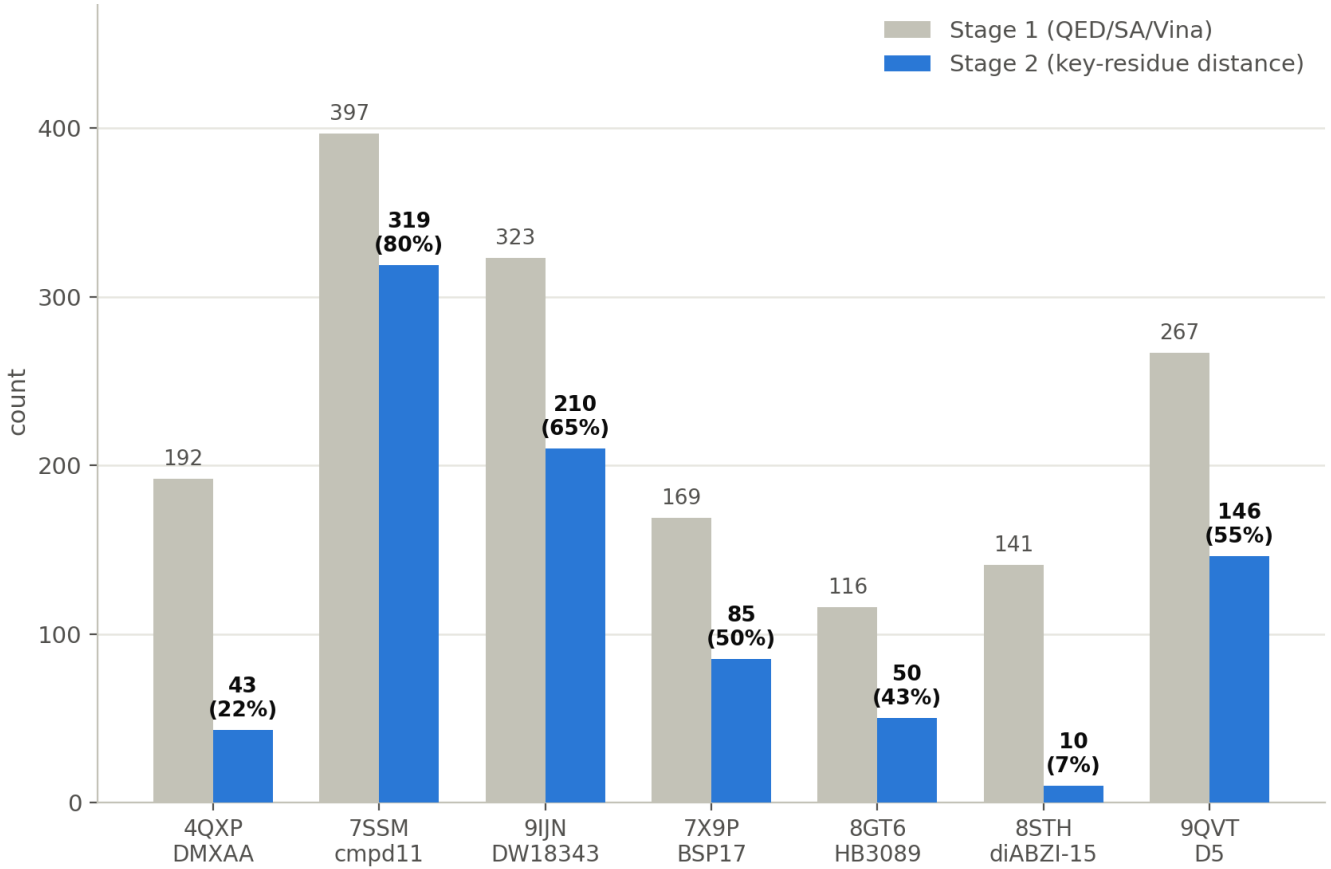
[Open conformation] (e.g. 8STH)



Contact with S162 and T263

Stage 2: Key residue contact Filter

863 candidates pass (ligand – key residue distance should be within crystal distance + margin)



8GT6 (43%), 8STH (7%)

Fragmentation
High MW (~490)
Narrow contact margin (Ser162)

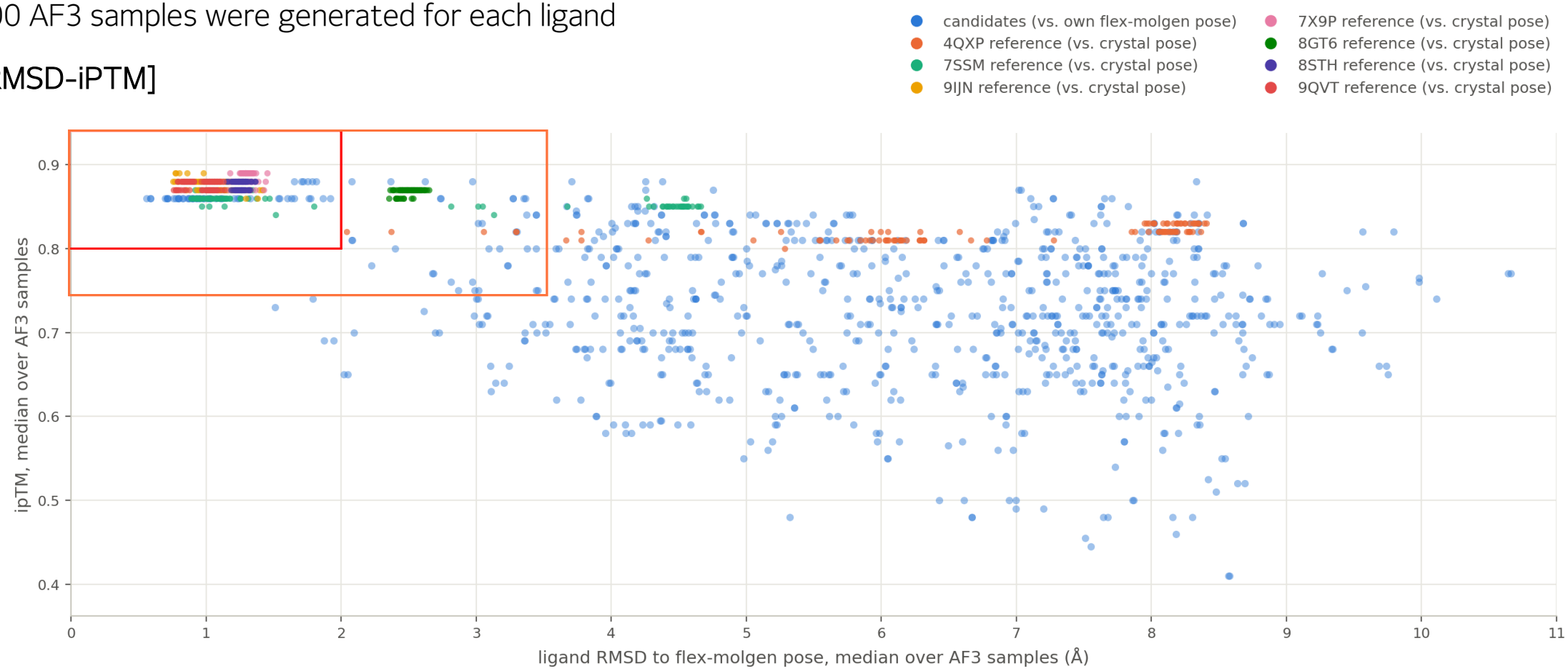
4QXP (22%)

Mutant-specific agonist
(not applicable with generation model)

Stage 3: AF3 reproducibility, consistency

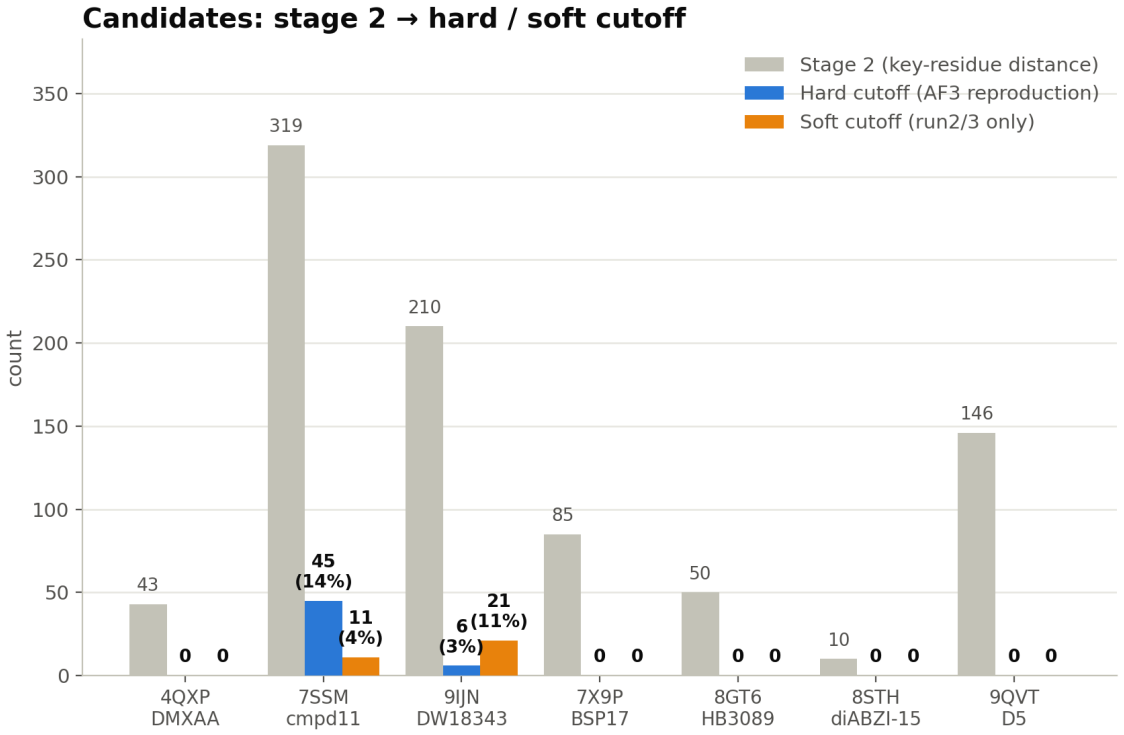
100 AF3 samples were generated for each ligand

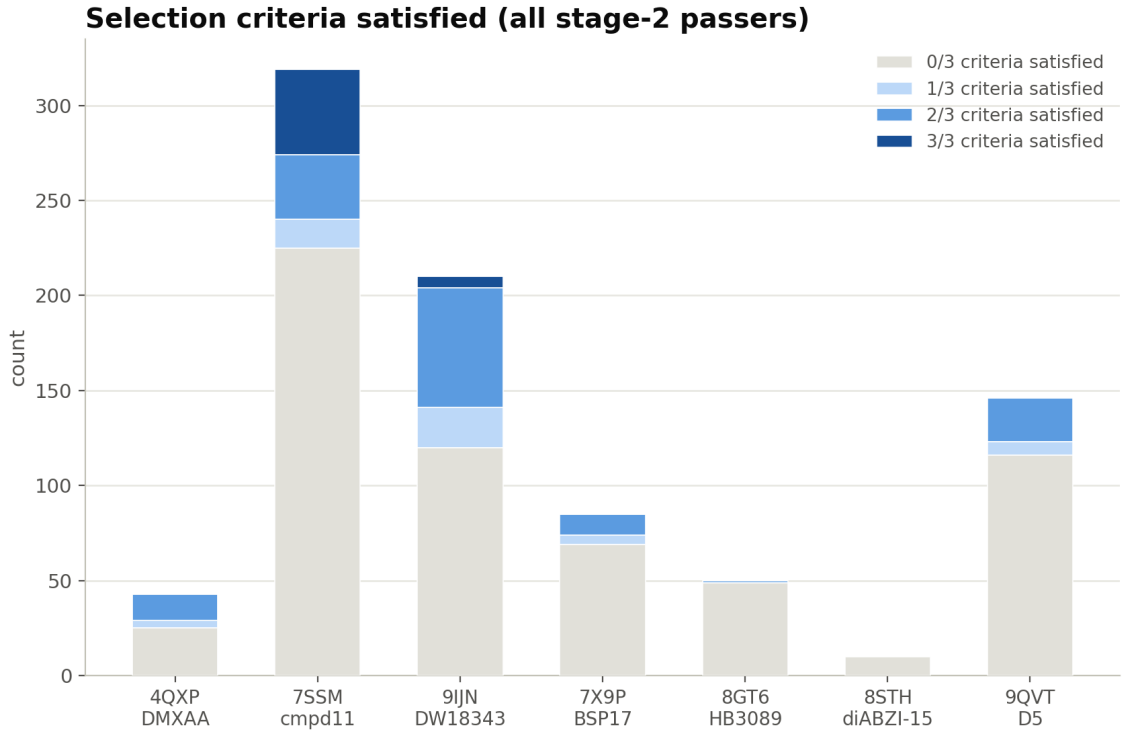
[RMSD-iPTM]



Stage 3: AF3 reproducibility, consistency

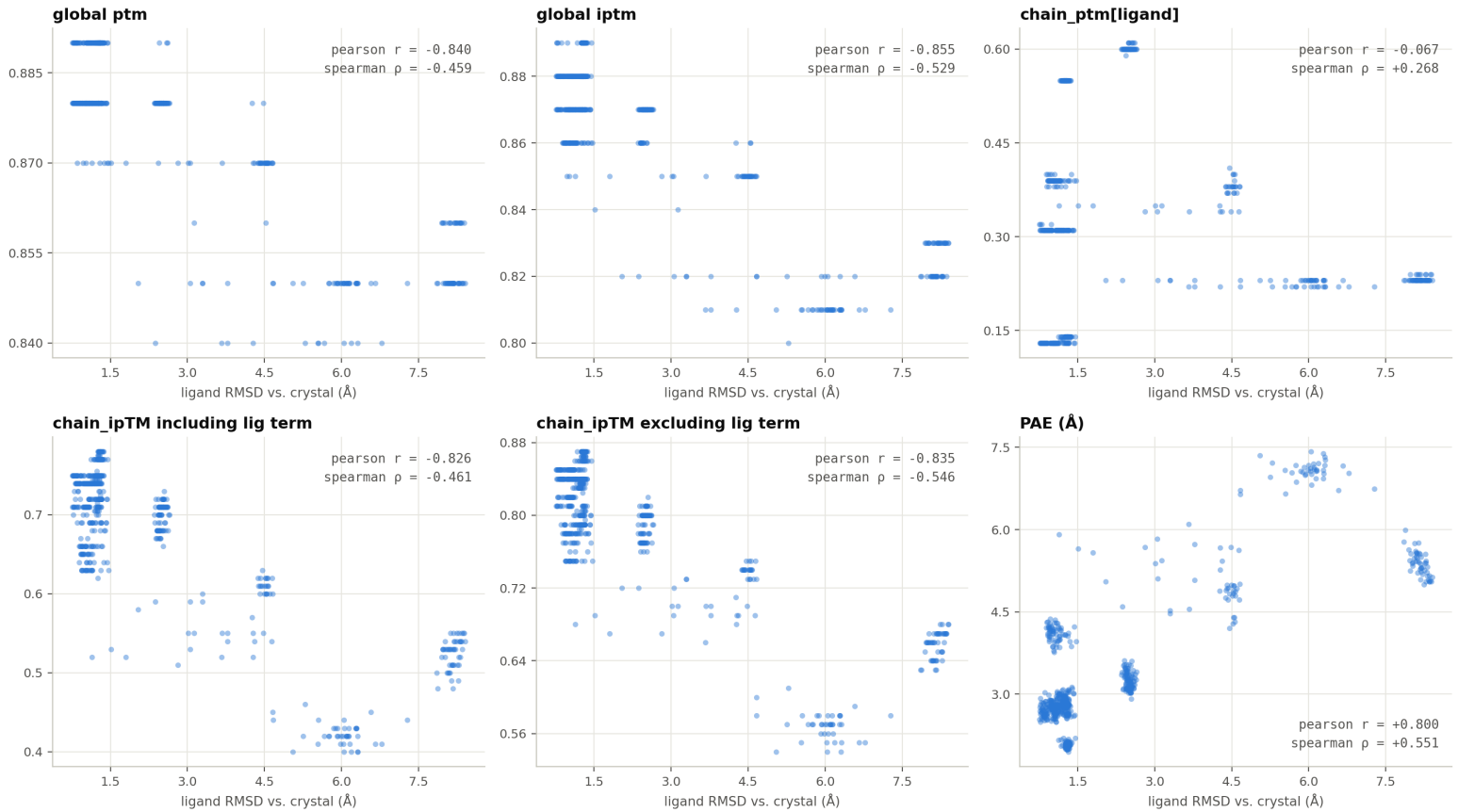
83 candidates passed (RMSD ≤ 3.5 Å, pTM ≥ 0.8, ipTM ≥ 0.75, all criteria were applied using the median)





Question 1

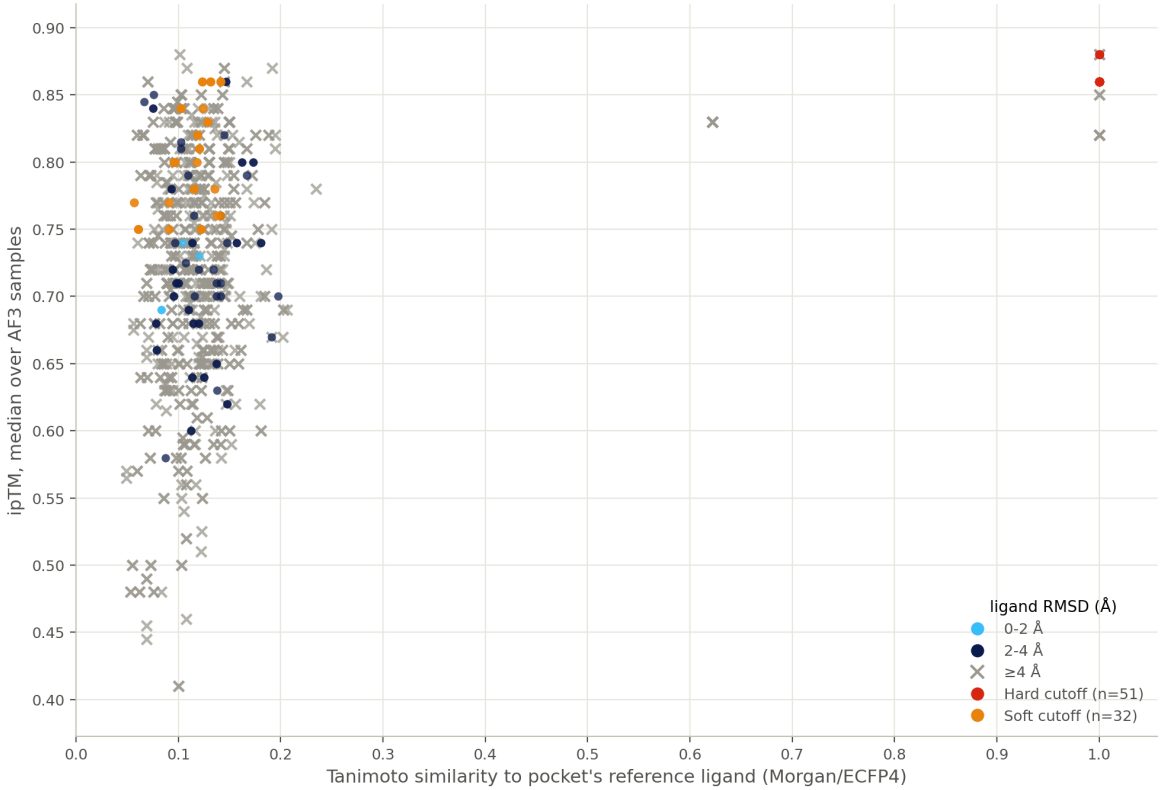
Which AF3 confidence metric best predicts ligand-pose RMSD?



Metric	Pearson r
Global pTM	- 0.840
Global ipTM	- 0.855
Chain pTM	- 0.067
Chain ipTM (incl. ligand chain)	- 0.826
Chain ipTM (excl. ligand chain)	- 0.835
PAE	+ 0.800

Question 2

Ligand novelty - AF3 confidence Tradeoff



Novel candidates were obtained when a looser cutoff was applied

Private data

Hard cutoff

RMSD ≤ 2.0 Å
pTM ≥ 0.85
ipTM ≥ 0.8

Soft cutoff

RMSD ≤ 3.5 Å
pTM ≥ 0.8
ipTM ≥ 0.75

Final candidates overview

Of the 32 candidates passing the soft cutoff, 20 were chemically unique molecules (remains were duplicate)

Private data

7SSM (cmpd11)

11 candidates
→ 8 unique molecules

9IJN (DW18343)

21 candidates
→ 12 unique molecules

Discussion / Future Plan

[Discussion]

- Ligand novelty와 AF3 confidence score의 적절한 Tradeoff 지점?
- Agonist vs Antagonist: 어떻게 하면 agonist만을 filtering 할 수 있을지?
- 적절한 Reference ligand selection을 위해 고려해야 할 요소들은 무엇이 있을지?

[Future Plan]

- Validation pipeline application: A008 (STING allosteric modulator), ACT1
- Final candidate screening (MD simulation, Retrosynthesis analysis - AiZynthFinder)